AI Tools for R

Day 1 - Introduction to Data Analysis with R

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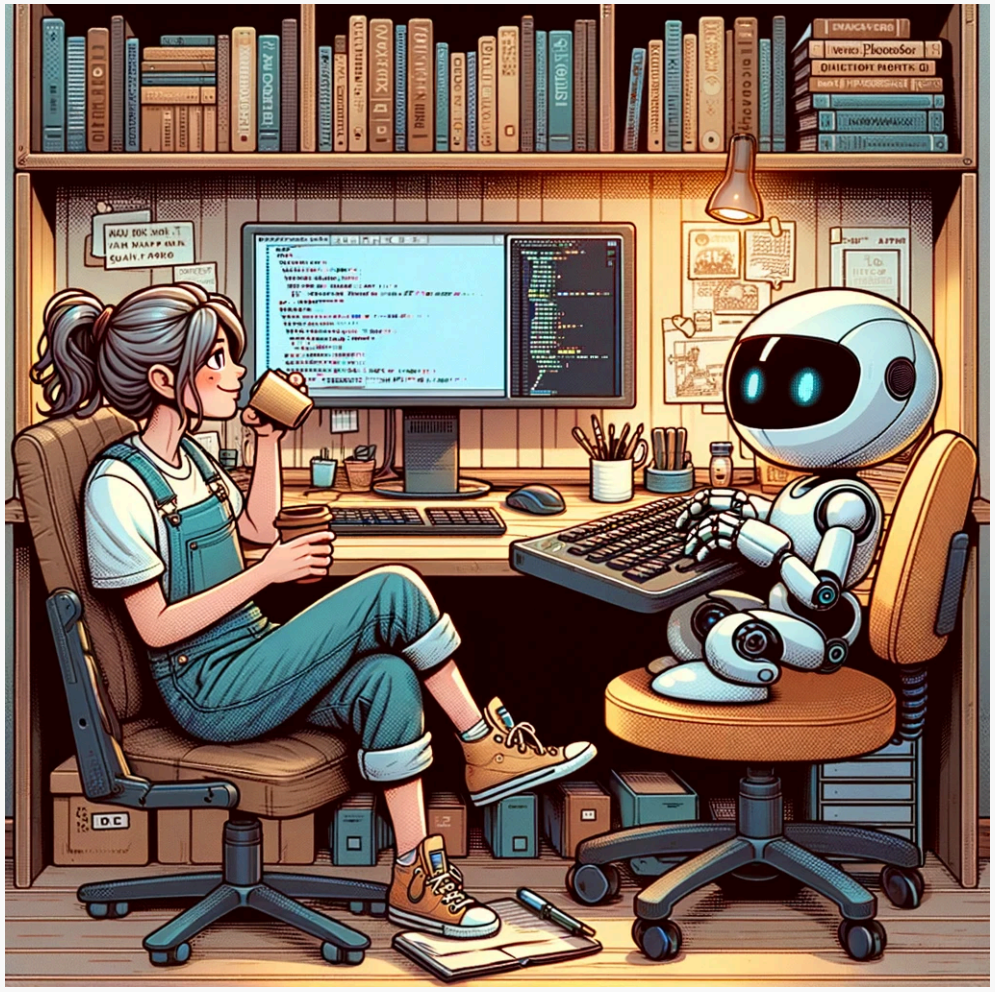
Motivation

- AI tools assist programmers with
  - Coding
  - Debugging
  - Learning
  - ...
- Higher productivity and efficiency
- More motivation
- But careful: You still need to understand what's going on!



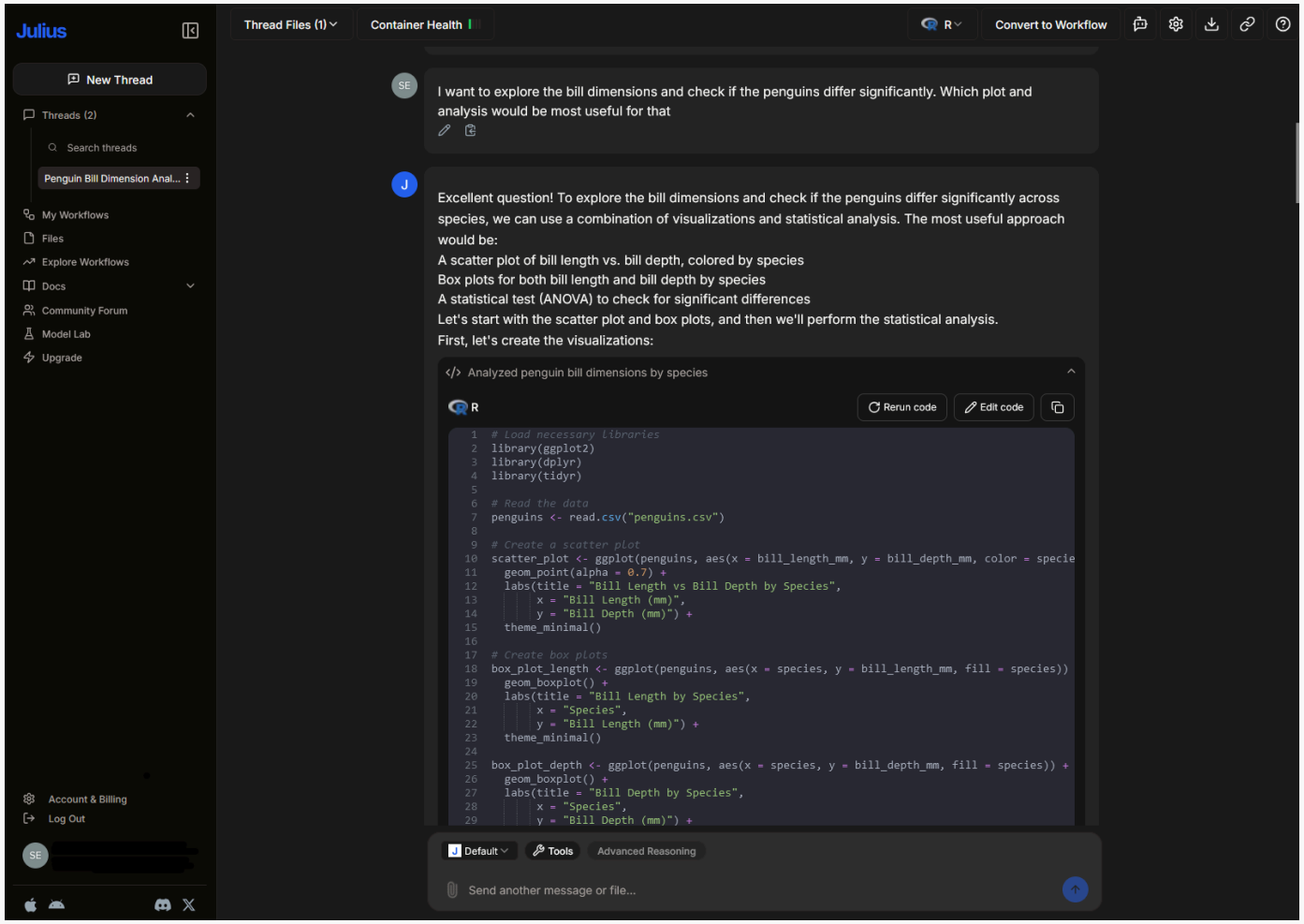
Overview of tools

- **Browser-based chat bots** (ChatGPT, Claude, ...)
  - General-purpose
- **Data-analysis tools** (Julius AI, RTutor, ...)
  - Upload data and ask questions about it
  - Can also execute code
  - Chat with your data
- **Integrated AI tools** (GitHub Copilot, ...)
  - Integrated directly in programming environment
  - Real-time suggestions, chat, debugging, ...

Find the tools that best fit your workflow!

Julius AI

- https://julius.ai/
- Try for free
- Basic plan ~20€ per months (-50% academic discount)
- Upload data and ask questions about it



Github Copilot

- https://github.com/features/copilot
- Based on models specifically trained on source code
- Basic idea: Integrate directly into your IDE
  - Real-time code suggestions (inline as you type)
  - Chat with the AI
- Works best for well-represented languages (Python, JS, ...), but R is also pretty good

Inline code suggestions

Available for RStudio and Positron

- Copilot tries to predict what you want to do next
- Suggestions are based on the context
  - Previous code
  - Comments
  - Variable and function names
  - …

fibonacci.R > fibonacci

```r
fibonacci <- function(n) {
  if (n == 0) {
    return(0)
  } else if (n == 1) {
    return(1)
  } else {
    return(fibonacci(n - 1) + fibonacci(n - 2))
  }
}
```

Get better suggestions

- Provide context
  - Open other files
  - Add top level comments explaining the purpose of the script
  - Name variables and functions properly
  - Copy-paste sample code and delete it later
- Be consistent
  - "Garbage in, garbage out"
  - Have a nice and consistent coding style

Nice side effect of using Copilot: More good-practice coding

Chat with GH Copilot about your

Only available in Positron or VS Code

- Open a chat window
- The AI can access you files and projects
- The AI can make taylored suggestions based on your project
- Great for
  - Debugging
  - Getting explanations
  - Getting good-practice suggestions
  - ...

How to get GitHub Copilot

See this website for step-by-step guide and more information.

It's really easy, but you need:

- GitHub Account
- Active GH Copilot subscription (10$ per month)
  - Get it for free as an academic with an educational account
- IDE that supports Copilot

Things to consider I

- Privacy
  - Check privacy guidelines before you use tools (e.g. `Github -> Seetings -> Copilot -> Policies`)
- Plagiarism
  - Block suggestions matching public code (`Github -> Seetings -> Copilot -> Policies`)

Things to consider II

- Responsibility
  - You are responsible for your scientific output
  - Stay critical, double-check
- Transparency
  - Make clear for which tasks you used which AI
- Know relevant guidelines
  - Journals
  - Your university
- Still understand what is happening!